Supplementary Material for: Standardised Comparison of Convolutional and Transformer Backbones for Brain Tumour Magnetic Resonance Image Classification under Perceptual-Hash Deduplication

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ABSTRACT

This document supplements the paper named in its title. It holds the dataset overview and sample images (Section S1), the deduplication and splitting details (Section S2), the complete training settings and evaluation procedures (Section S3), the training curves and per-run results (Section S4), the fold-ensemble test results of binary detection and subtype classification (Section S5), the calibration and efficiency details (Section S6), and the interpretability and shortcut details (Section S7). Sections, figures and tables numbered with an S belong to this document; those without an S belong to the paper.

S1. Data

Images per source and class and the native image sizes of both pools, drawn from the collections of Chakrabarty (2019), Hamada (2025), Bhuvaji et al. (2020) and Cheng (2017), are shown in Figure S1. The binary detection sources span 184 and 291 distinct image sizes, whereas 99.5% of the Cheng slices and 93.1% of the Bhuvaji training folder are 512×512 pixels; the Bhuvaji testing folder is far less uniform, with 184 distinct sizes.

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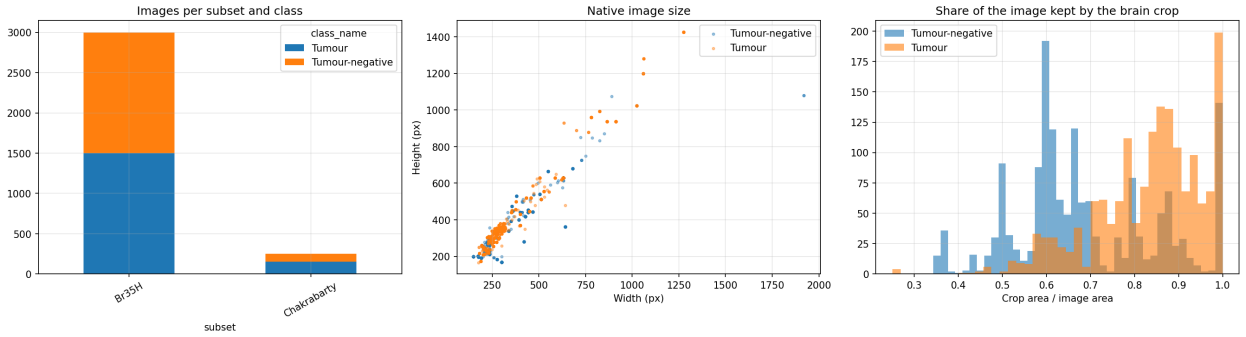
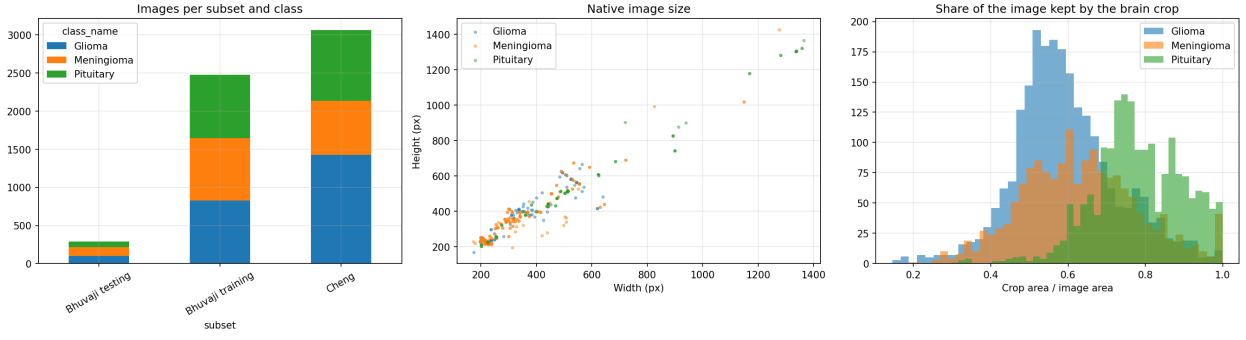


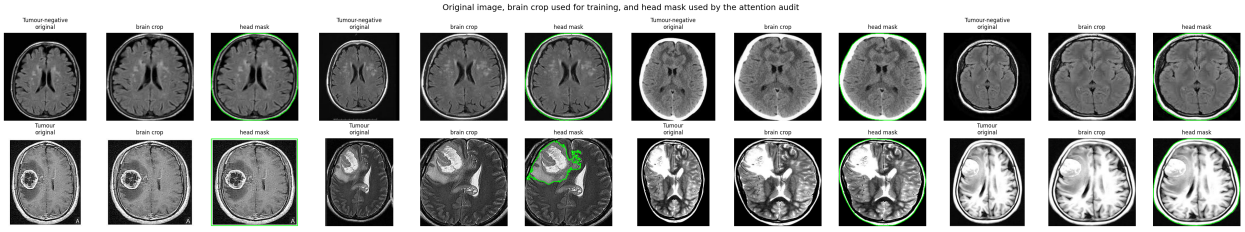
(a) Binary detection.



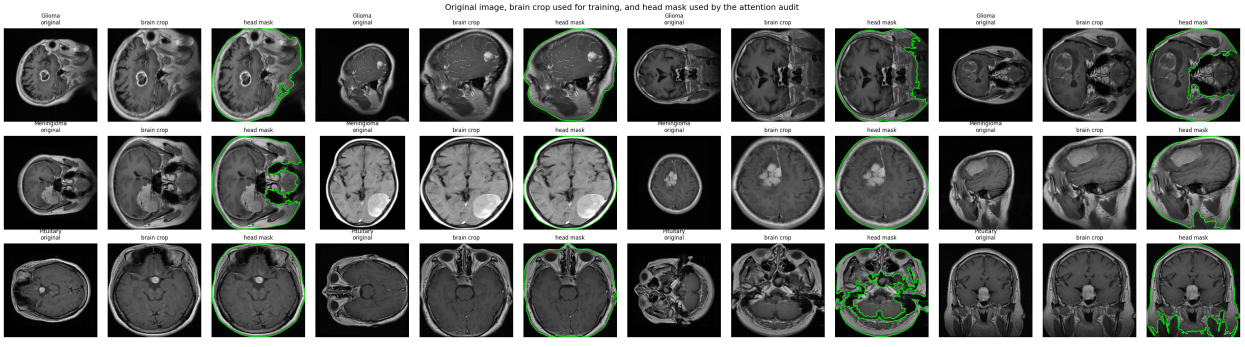
(b) Subtype classification.

Figure S1: Dataset overview before deduplication: images per source and class, and native image sizes.

The MRI sequence is unknown in both pools. Neither binary detection source records the sequence (T1-weighted, contrast-enhanced T1-weighted, T2-weighted or FLAIR) of individual images. In the subtype classification pool, the Cheng component is contrast-enhanced T1-weighted according to its source publication (Cheng et al., 2015), but the Bhuvaji collection records no sequence information, so the pooled set is a mixture of unknown composition, and a model trained on it cannot be specified for deployment on any particular acquisition protocol. Representative images, their brain crops and the head masks used by the attention and shortcut analyses are shown in Figure S2.



(a) Binary detection.



(b) Subtype classification.

Figure S2: Sample images of each class, the brain crop that every model is trained and evaluated on, and the head masks used by the attention and shortcut analyses.

S2. Deduplication and Splitting

Duplicates were removed keeping the copy from the first-listed folder. In the binary detection pool, of the 2,068 removals, 729 were Br35H images whose retained duplicate is a Chakrabarty image, 1,292 were duplicates within Br35H and 47 within Chakrabarty. In the subtype classification pool, 31 cross-label pairs lay within the tolerance, all glioma against meningioma: 8 pairs inside the Bhuvaji collection had a normalised cross-correlation (NCC) (Zitová and Flusser, 2003) of at least 0.9997 and were excluded as label noise, and 23 were hash false positives with NCC between 0.67 and 0.86. Deduplication removed 1,094 subtype classification images as duplicates, 608 within Bhuvaji, 483 within Cheng and only 3 Cheng meningioma images against a Bhuvaji copy, and the label-noise exclusion brought the pool to 4,730 images. Per image, the split table records 1,088 duplicates and 10 label-noise exclusions (2 glioma, 8 meningioma); both accounts reach the same 4,730 images. Images kept at other tolerances are listed in Table S1; tolerances 4 and 5 give the same pool in both tasks. The cross-label pairs are shown in Figure S3.

Table S1
Images kept by the label-aware perceptual-hash pass at each Hamming tolerance. The study uses tolerance 5. The subtype classification count at tolerance 5 is before the label-noise exclusion, which brings the pool to 4,730.

Tolerance	0, 1	2, 3	4, 5	6, 7	8, 9	10, 11	12
Binary detection							
Tumour-negative	860	583	455	395	341	310	285
Tumour	1,217	912	730	607	541	494	424
Total	**2,077**	**1,495**	**1,185**	**1,002**	**882**	**804**	**709**
Subtype classification							
Glioma	2,303	2,221	2,010	1,653	1,277	919	627
Meningioma	1,518	1,453	1,316	1,130	935	704	506
Pituitary	1,751	1,663	1,408	1,145	886	663	478
Total	**5,572**	**5,337**	**4,734**	**3,928**	**3,098**	**2,286**	**1,611**

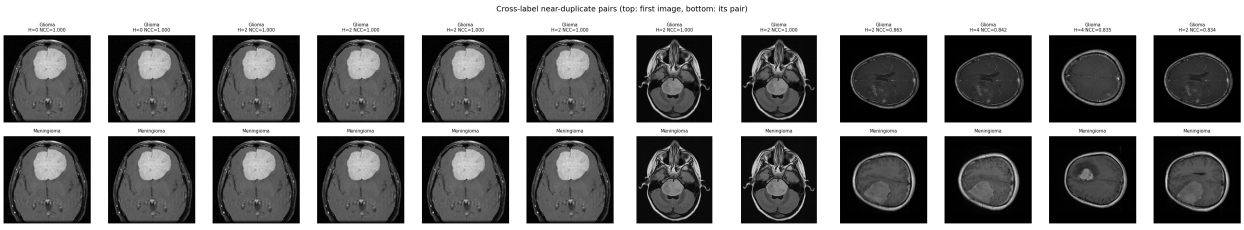
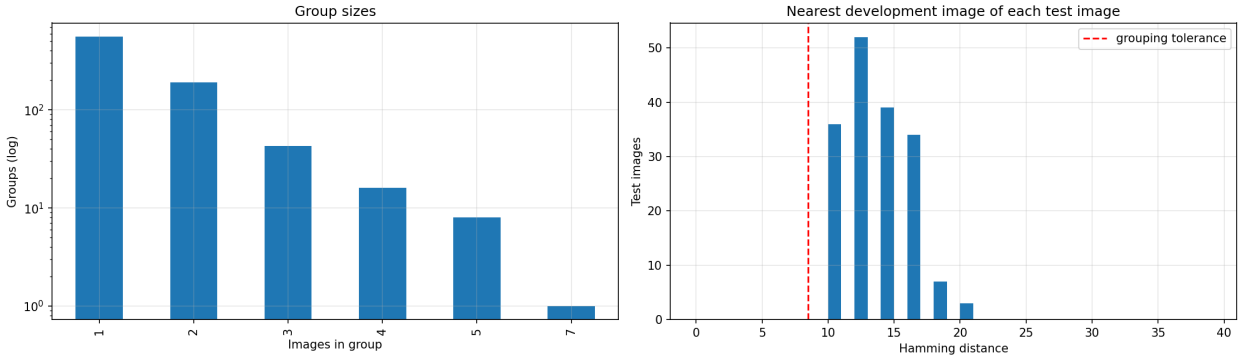
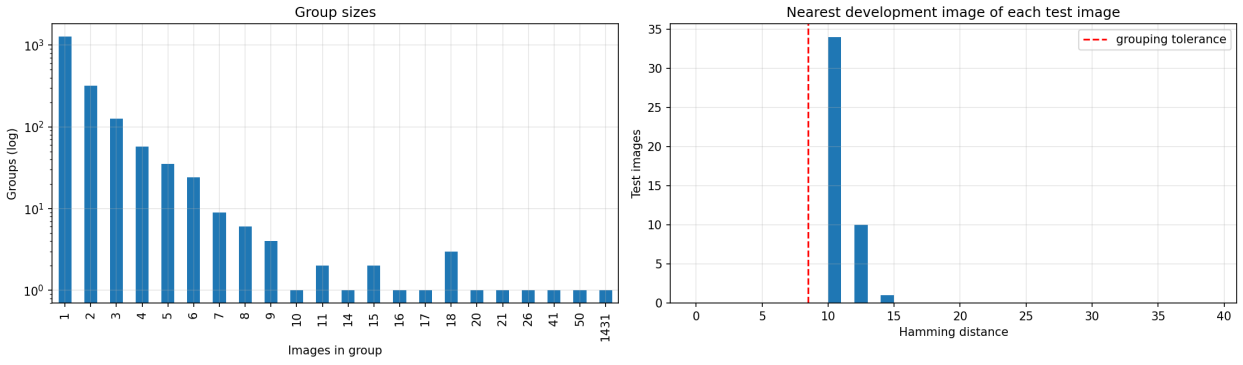


Figure S3: Subtype classification cross-label pairs within Hamming distance 5, label-noise pairs first. All 31 pairs are glioma against meningioma; the 8 pairs at NCC $\geq$ 0.98 lie inside the Bhuvaji collection and both copies were excluded, and the 23 others are hash false positives and were kept. The binary detection pool has no cross-label pair.

The leakage audit measured, for every test image, the nearest development image by Hamming distance and NCC (Figure S4). The subtype classification grouping produced one group of 1,431 images, more than 5% of the pool, and the notebook warned that a smaller grouping tolerance should be considered; the run kept the tolerance of 8.



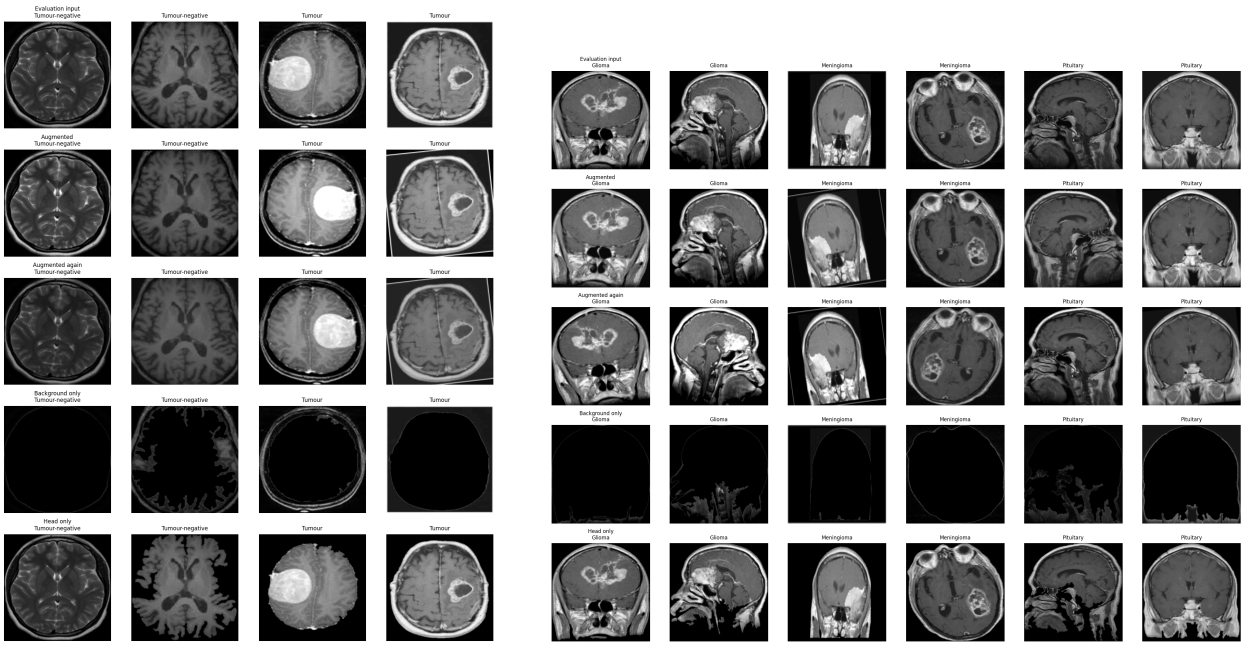
(a) Binary detection: 821 groups, the largest of 7 images.



(b) Subtype classification: 1,867 groups, the largest of 1,431 images.

Figure S4: Group sizes and the distance from each test image to its nearest development image. No test image has a development image within Hamming distance 5 or 8 or at NCC $\geq$ 0.98.

Crops are scaled to [0, 1] and normalised with the ImageNet (Deng et al., 2009) channel statistics: means of 0.485, 0.456 and 0.406 and standard deviations of 0.229, 0.224 and 0.225 for the red, green and blue channels. Training augmentation runs on the GPU and consists of horizontal flipping with probability 0.5, rotation of up to $\pm$10 degrees, and brightness and contrast jitter of up to 20%; validation and test images are not augmented (Figure S5).



(a) Binary detection. (b) Subtype classification.

Figure S5: Training augmentation (horizontal flip, rotation up to $\pm 10°$, brightness and contrast jitter of 20%) on a few training images, with the background-only and head-only inputs of the masking and shortcut tests.

S3. Training and Evaluation Details

All experiments ran on Kaggle with two NVIDIA Tesla T4 graphics processing units, four central processing unit cores and 33.7 GB of system memory, using Python 3.13.15, PyTorch 2.11.0 (Paszke et al., 2019), torchvision 0.26.0, timm 1.0.29 and CUDA 12.8. Two worker processes, one per GPU, trained different (model, fold) jobs in parallel. The settings in Table S2 were identical for every model and both tasks.

Table S2
Training settings, identical for all nine models and both tasks.

Setting	Value
Epochs	20, with 1 warm-up epoch
Effective batch	64 images, through gradient accumulation
Optimiser steps	Equal for every model on a given fold (binary detection 260; subtype classification 780 to 1,420 by fold)
Optimiser	AdamW (Loshchilov and Hutter, 2019), learning rate 3×10^{-4}, weight decay 0.01
Learning rate schedule	Linear warm-up, then cosine decay per optimiser step (Loshchilov and Hutter, 2017)
Loss	Binary cross-entropy with logits (binary detection); cross-entropy (subtype classification)
Precision	Half-precision autocasting with gradient scaling (Micikevicius et al., 2018)
Gradient clipping	Norm 1.0 (Pascanu et al., 2013)
Input resolution	224×224 pixels; 299×299 pixels for InceptionV3
Augmentation	Horizontal flip ($p = 0.5$), rotation up to $\pm 10°$, brightness and contrast jitter of 20%
Checkpoint selection	Lowest validation log-loss
Cross-validation	5 group-aware folds (Stone, 1974; Pedregosa et al., 2011) plus a fixed held-out test set
Random seed	42

The largest micro-batch that fitted on one GPU was probed for each model, and gradient accumulation made up the difference: Swin-Base trained with micro-batches of 32 accumulated over two steps, every other model with micro-batches of 64. All models therefore receive the same number of optimiser steps on the same fold: 260 steps for every binary detection run (13 steps per epoch), and 1,420, 1,240, 1,260, 1,200 and 780 steps on subtype classification folds 0 to 4, whose training sets differ in size because of the uneven split. Every training image is used in every epoch. After every epoch the validation fold was evaluated, and the checkpoint with the lowest validation log-loss, a strictly proper scoring rule (Gneiting and Raftery, 2007), was kept. With 45 full-image runs and 9 background-only runs per task, each notebook trained 54 jobs; all 54 finished in both tasks, in 63.5 minutes of training for binary detection and 264.6 minutes for subtype classification. Deterministic kernels were not enforced, so re-running the notebooks should reproduce the reported figures closely but not necessarily bit for bit.

The temperature search (Guo et al., 2017) covered $\log T$ in $[-3, 3]$. Binary detection sensitivity and specificity are reported at a threshold of 0.5 and at the mean validation-tuned threshold. For each model, the bootstrap (Efron, 1979) 95% confidence interval of its accuracy gap to the leading model was also computed; a model whose interval includes zero cannot be separated from the leader on this test set. Test accuracy was further broken down by the Hamming distance from each test image to its nearest development image. Grad-CAM (Selvaraju et al., 2017) was computed for every test image at the last spatial or token stage of the network, which for VGG16 is the last convolution, on the fold model with the lowest validation loss. Because Grad-CAM maps of convolutional and attention stages are not strictly commensurable, occlusion sensitivity (Zeiler and Fergus, 2014) was computed in the same way for every model on a common stratified sample of test images. For efficiency, single-image latency was measured over 100 passes after 10 warm-up passes, and throughput at batch size 32 over 20 passes after 3 warm-up passes, with the whole protocol repeated three times; training time, training throughput and peak training memory are averaged over the folds.

S4. Cross-Validation Details

The training curves of every binary detection model are shown in Figure S6.

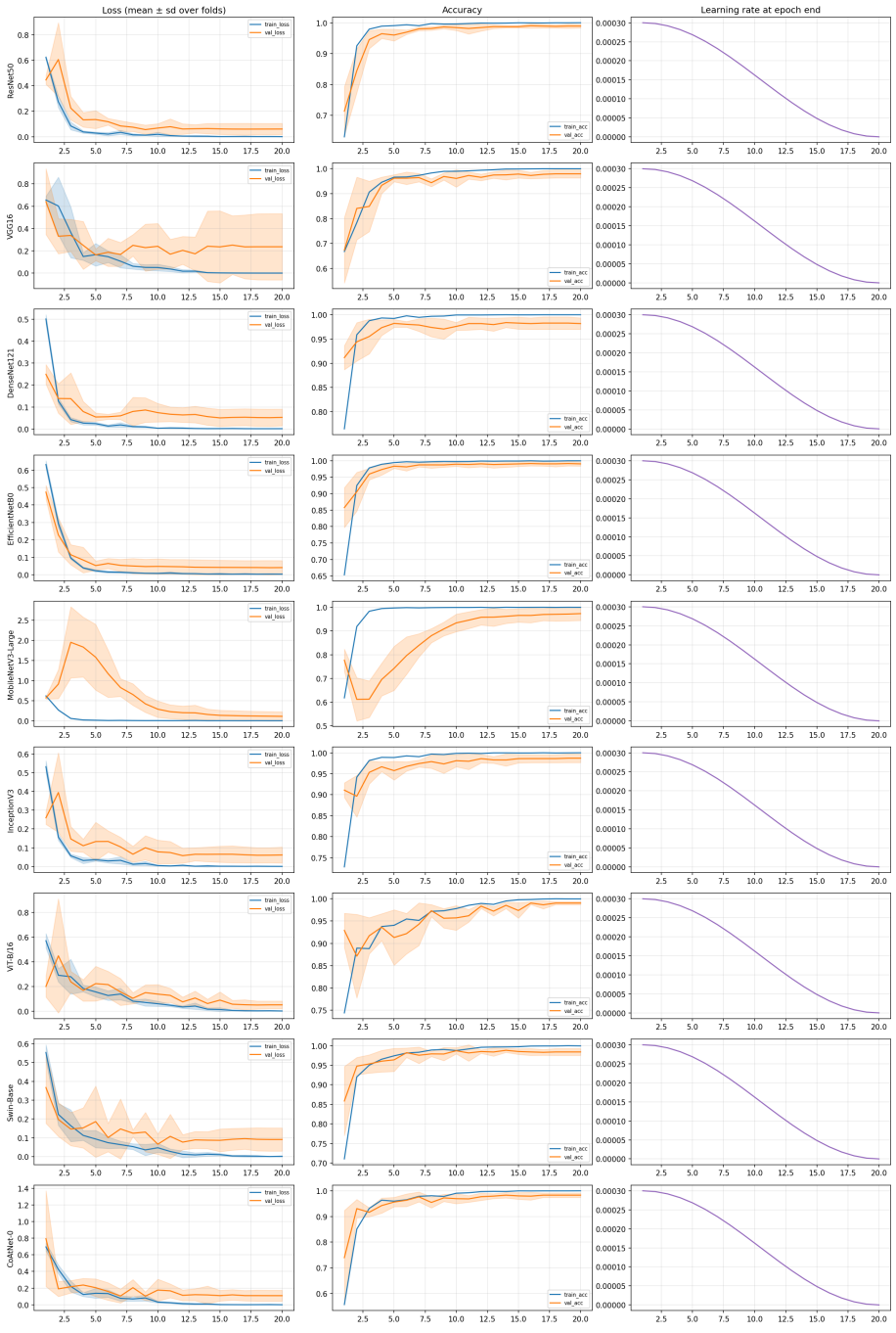


Figure S6: Binary detection: mean ± one standard deviation across folds of training and validation loss, validation accuracy and learning rate, per model.

The same curves for every subtype classification model are shown in Figure S7.

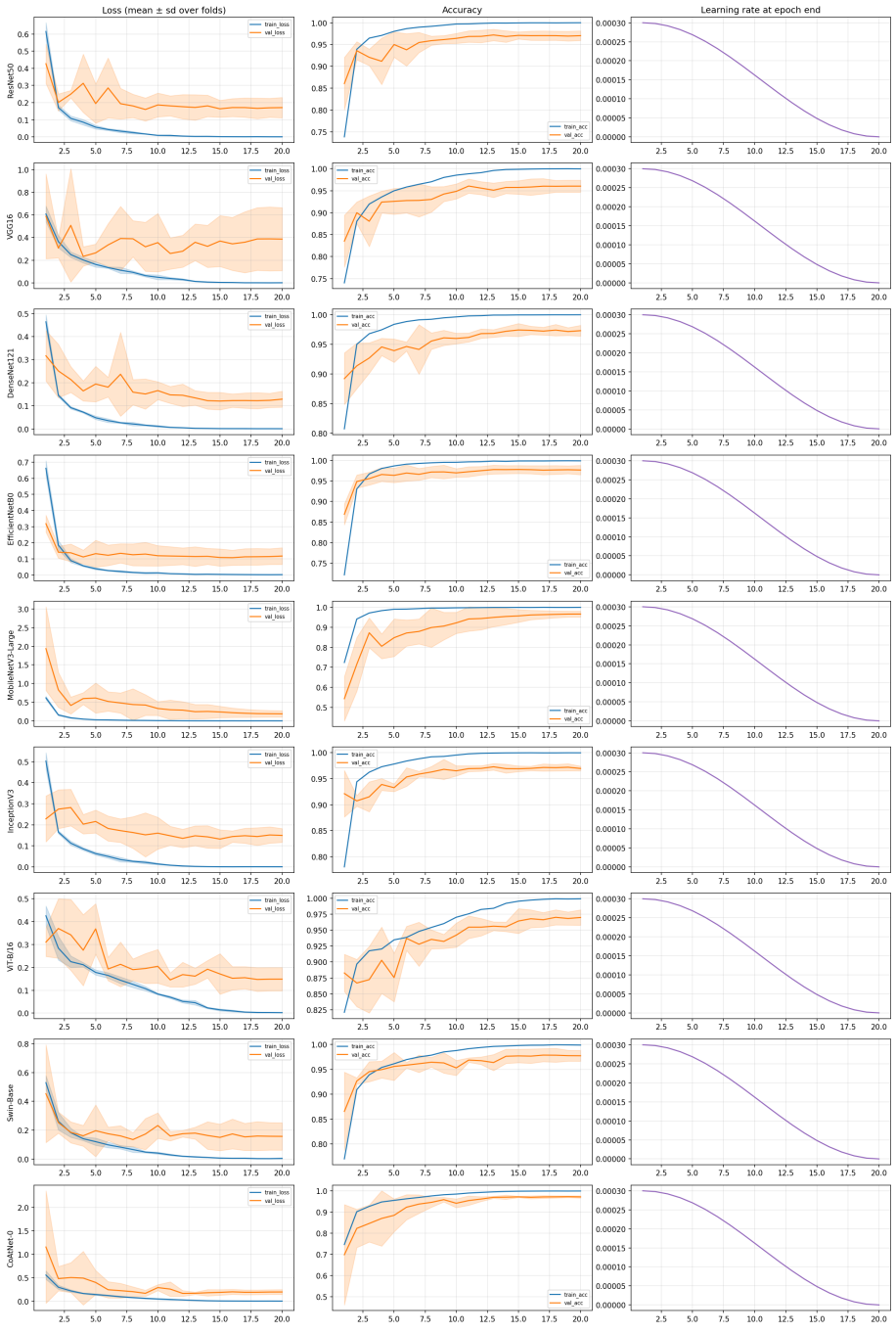


Figure S7: Subtype classification: mean ± one standard deviation across folds of training and validation loss, validation accuracy and learning rate, per model.

The fitted temperature (Guo et al., 2017), the Youden threshold (Youden, 1950) and the test result of every binary detection run are listed in Table S3.

Table S3
Binary detection: every full-image run. Every run took 260 optimiser steps. T is the temperature fitted on the run's validation fold and Thr. the Youden threshold chosen there; accuracy and log-loss are on the 171 test images after temperature scaling. In fold 4 the temperature of six models lies at or near the lower search bound $e^{-3} \approx 0.0498$.

Model	Fold	Best epoch	T	Thr.	Test accuracy	Test log-loss	Train images
CoAtNet-0	0	9	1.1767	0.739	0.9883	0.0491	830
	1	7	1.4761	0.529	0.9825	0.0534	803
	2	12	1.7756	0.633	0.9474	0.1317	807
	3	14	2.1222	0.316	0.9883	0.0333	804
	4	7	0.0604	1.000	0.9708	0.8079	812
DenseNet121	0	6	0.8003	0.724	0.9766	0.0847	830
	1	11	1.0106	0.657	0.9649	0.0751	803
	2	12	1.3466	0.885	0.9649	0.1019	807
	3	14	1.1351	0.950	0.9825	0.0696	804
	4	15	0.5882	0.927	0.9708	0.2513	812
EfficientNetB0	0	5	1.2527	0.621	0.9708	0.0875	830
	1	14	1.2311	0.962	0.9766	0.0378	803
	2	13	1.1739	0.625	0.9825	0.0471	807
	3	11	0.9215	0.985	0.9825	0.0755	804
	4	19	0.0498	1.000	0.9766	0.5431	812
InceptionV3	0	17	1.6119	0.895	0.9766	0.0543	830
	1	8	1.4656	0.045	0.9766	0.0603	803
	2	10	0.9470	0.861	0.9766	0.1005	807
	3	5	1.1069	0.829	0.9825	0.0372	804
	4	12	0.0682	1.000	0.9825	0.3662	812
MobileNetV3-Large	0	20	2.7638	0.121	0.9298	0.1551	830
	1	20	1.8160	0.583	0.9766	0.0638	803
	2	14	1.6694	0.410	0.9883	0.0448	807
	3	20	1.7832	0.249	0.9591	0.1161	804
	4	13	0.0508	1.000	0.9649	0.7224	812
ResNet50	0	9	1.7422	0.819	0.9474	0.1451	830
	1	8	1.7905	0.314	0.9766	0.0837	803
	2	7	1.0362	0.853	0.9649	0.0787	807
	3	17	1.2096	0.759	0.9708	0.1416	804
	4	12	0.0498	1.000	0.9825	0.4848	812
Swin-Base	0	14	1.7903	0.617	0.9708	0.0826	830
	1	11	1.7026	0.550	0.9708	0.1311	803
	2	6	1.5159	0.683	0.9591	0.1221	807
	3	5	0.8935	0.261	0.9883	0.0536	804
	4	10	0.0498	1.000	1.0000	0.0000	812
VGG16	0	5	1.7926	0.462	0.9532	0.1252	830
	1	13	2.1367	0.216	0.9649	0.0912	803
	2	18	1.5436	0.775	0.9649	0.2265	807
	3	4	1.4973	0.196	0.9474	0.1276	804
	4	6	1.8620	0.495	0.9649	0.1407	812
ViT-B/16	0	17	1.4451	0.767	0.9825	0.0566	830
	1	12	1.5874	0.483	0.9649	0.1114	803
	2	17	1.3066	0.824	0.9883	0.0526	807
	3	8	0.8068	0.559	0.9708	0.0711	804
	4	16	0.9993	0.836	0.9708	0.1007	812

The fitted temperature, the number of optimiser steps and the test result of every subtype classification run are listed in Table S4.

Table S4
Subtype classification: every full-image run. Optimiser steps depend on the fold because the group-aware folds differ in size; within a fold they are equal for every model. In fold 0, whose validation fold has 154 images, the temperature of EfficientNetB0, InceptionV3 and Swin-Base lies at or near the lower search bound.

Model	Fold	Steps	Best epoch	T	Test accuracy	Test log-loss	Train / val images
CoAtNet-0	0	1,420	7	0.8899	1.0000	0.0209	4,531 / 154
	1	1,240	9	1.3042	1.0000	0.0034	3,943 / 742
	2	1,260	5	0.8503	0.9778	0.0720	3,988 / 697
	3	1,200	6	1.2193	0.9778	0.0395	3,809 / 876
	4	780	15	2.3073	0.9778	0.0236	2,469 / 2,216
DenseNet121	0	1,420	7	1.1754	1.0000	0.0040	4,531 / 154
	1	1,240	15	1.9404	1.0000	0.0061	3,943 / 742
	2	1,260	9	1.5745	1.0000	0.0186	3,988 / 697
	3	1,200	4	1.4594	0.9778	0.1067	3,809 / 876
	4	780	10	1.5314	0.9778	0.0355	2,469 / 2,216
EfficientNetB0	0	1,420	9	0.0498	0.9778	0.6140	4,531 / 154
	1	1,240	4	1.5348	0.9111	0.1585	3,943 / 742
	2	1,260	15	1.8186	1.0000	0.0024	3,988 / 697
	3	1,200	3	1.3621	1.0000	0.0181	3,809 / 876
	4	780	16	1.7429	0.9778	0.0330	2,469 / 2,216
InceptionV3	0	1,420	9	0.0515	1.0000	0.0000	4,531 / 154
	1	1,240	7	1.5646	1.0000	0.0050	3,943 / 742
	2	1,260	10	1.6805	1.0000	0.0040	3,988 / 697
	3	1,200	15	2.2675	1.0000	0.0101	3,809 / 876
	4	780	11	1.7383	0.9778	0.0503	2,469 / 2,216
MobileNetV3-Large	0	1,420	12	1.0519	0.9778	0.1369	4,531 / 154
	1	1,240	20	3.1578	0.9778	0.0712	3,943 / 742
	2	1,260	18	2.6818	0.9778	0.0685	3,988 / 697
	3	1,200	8	2.4891	0.9778	0.0327	3,809 / 876
	4	780	20	3.0036	0.9778	0.0689	2,469 / 2,216
ResNet50	0	1,420	5	0.9164	1.0000	0.0026	4,531 / 154
	1	1,240	9	2.0543	0.9778	0.0395	3,943 / 742
	2	1,260	5	1.5465	0.9778	0.0672	3,988 / 697
	3	1,200	15	2.6184	1.0000	0.0176	3,809 / 876
	4	780	8	1.6629	0.9778	0.0957	2,469 / 2,216
Swin-Base	0	1,420	15	0.0498	1.0000	0.0000	4,531 / 154
	1	1,240	8	1.6375	1.0000	0.0112	3,943 / 742
	2	1,260	9	1.9777	1.0000	0.0076	3,988 / 697
	3	1,200	5	1.7771	0.9778	0.0397	3,809 / 876
	4	780	18	2.2403	1.0000	0.0047	2,469 / 2,216
VGG16	0	1,420	11	1.2077	0.9778	0.1367	4,531 / 154
	1	1,240	7	1.7136	1.0000	0.0175	3,943 / 742
	2	1,260	2	1.6416	0.9333	0.2497	3,988 / 697
	3	1,200	4	2.0744	0.9778	0.0520	3,809 / 876
	4	780	13	2.2774	0.9778	0.0970	2,469 / 2,216
ViT-B/16	0	1,420	15	1.3910	1.0000	0.0020	4,531 / 154
	1	1,240	13	1.7649	1.0000	0.0238	3,943 / 742
	2	1,260	13	1.2801	0.9778	0.0420	3,988 / 697
	3	1,200	11	1.5805	1.0000	0.0129	3,809 / 876
	4	780	11	1.4512	0.9778	0.1176	2,469 / 2,216

S5. Fold-Ensemble Test Results

Table S5 gives the binary detection fold ensembles and Table S6 the subtype classification fold ensembles.

Table S5
Binary detection: fold ensembles on the 171 held-out test images (107 tumour, 64 tumour-negative). Sensitivity is recall of the tumour class and specificity recall of the tumour-negative class, both at threshold 0.5; FN and FP are missed tumours and false alarms. Accuracy has a bootstrap 95% confidence interval. Rows are ordered by accuracy.

Model	Accuracy (%) [95% CI]	Bal. acc. (%)	Sens. (%)	Spec. (%)	MCC	AUC	FN	FP
CoAtNet-0	99.42 [98.25, 100]	99.53	99.07	100.00	0.9876	0.9994	1	0
EfficientNetB0	99.42 [98.25, 100]	99.22	100.00	98.44	0.9876	0.9991	0	1
ViT-B/16	99.42 [98.25, 100]	99.22	100.00	98.44	0.9876	0.9984	0	1
Swin-Base	98.83 [97.08, 100]	99.07	98.13	100.00	0.9755	0.9997	2	0
InceptionV3	98.25 [95.91, 100]	97.97	99.07	96.88	0.9625	0.9991	1	2
ResNet50	98.25 [95.91, 100]	98.28	98.13	98.44	0.9627	0.9988	2	1
DenseNet121	97.08 [94.15, 99.42]	96.41	99.07	93.75	0.9376	0.9984	1	4
MobileNetV3-Large	97.08 [94.15, 99.42]	97.04	97.20	96.88	0.9378	0.9978	3	2
VGG16	97.08 [94.15, 99.42]	97.04	97.20	96.88	0.9378	0.9974	3	2

Table S6
Subtype classification: fold ensembles on the 45 held-out test images (1 glioma, 13 meningioma, 31 pituitary). Precision and F1 are macro-averaged; AUC is the macro one-versus-rest value. Accuracy has a bootstrap 95% confidence interval.

Model	Accuracy (%) [95% CI]	Bal. acc. (%)	Prec. (%)	F1 (%)	MCC	AUC
ResNet50	100.00 [100, 100]	100.00	100.00	100.00	1.0000	1.0000
DenseNet121	100.00 [100, 100]	100.00	100.00	100.00	1.0000	1.0000
InceptionV3	100.00 [100, 100]	100.00	100.00	100.00	1.0000	1.0000
ViT-B/16	100.00 [100, 100]	100.00	100.00	100.00	1.0000	1.0000
Swin-Base	100.00 [100, 100]	100.00	100.00	100.00	1.0000	1.0000
CoAtNet-0	100.00 [100, 100]	100.00	100.00	100.00	1.0000	1.0000
VGG16	97.78 [93.33, 100]	97.44	83.33	87.56	0.9515	1.0000
EfficientNetB0	97.78 [93.33, 100]	97.44	83.33	87.56	0.9515	1.0000
MobileNetV3-Large	97.78 [93.33, 100]	97.44	83.33	87.56	0.9515	1.0000

The confusion matrices of the binary detection ensembles are shown in Figure S8.

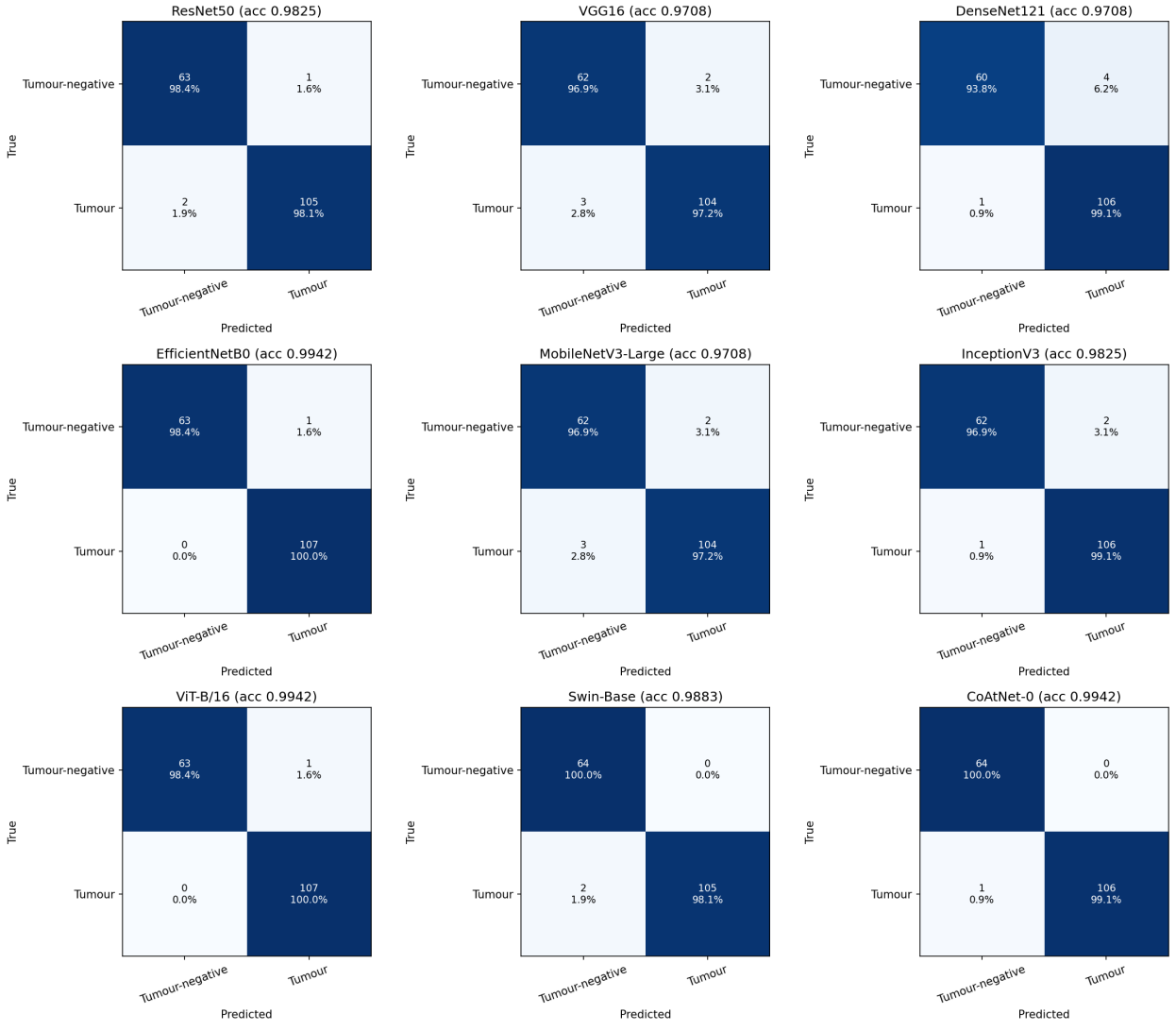


Figure S8: Binary detection: test confusion matrices of the fold ensemble of every model, as counts with row percentages.

The confusion matrices of the subtype classification ensembles are shown in Figure S9.

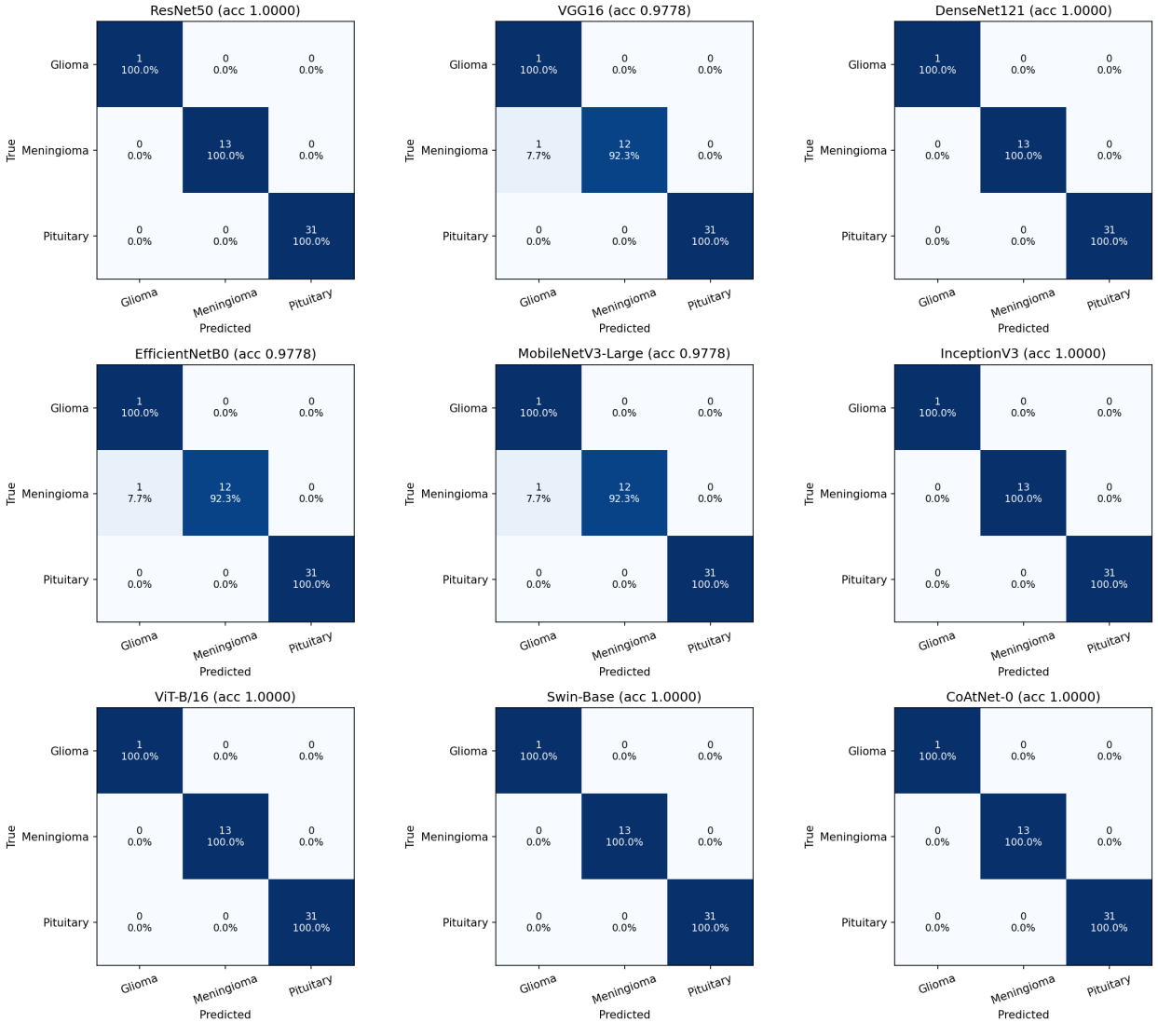
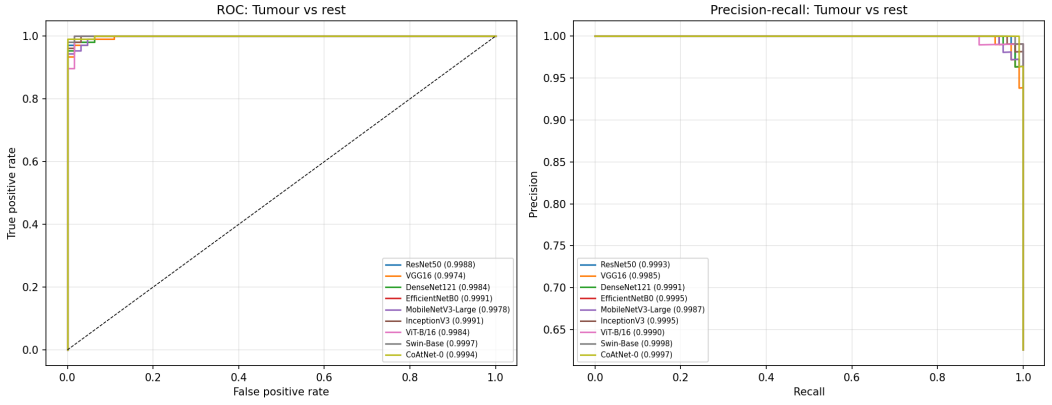
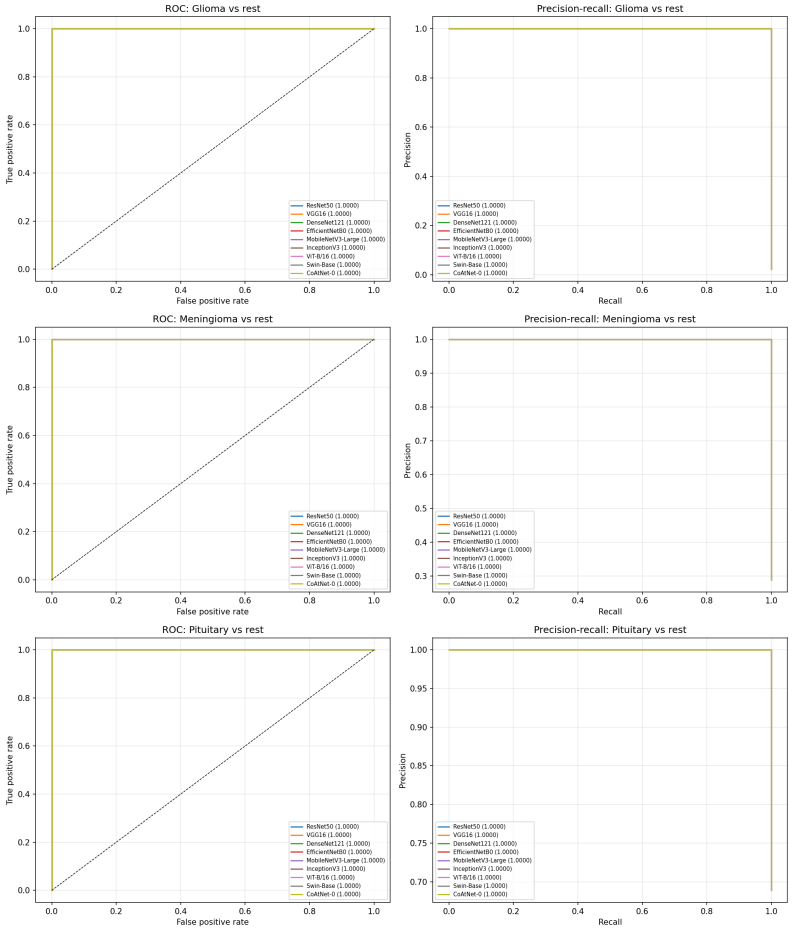


Figure S9: Subtype classification: test confusion matrices of every fold ensemble, as counts with row percentages.

The receiver operating characteristic and precision–recall curves (Fawcett, 2006) of every ensemble are shown in Figure S10.



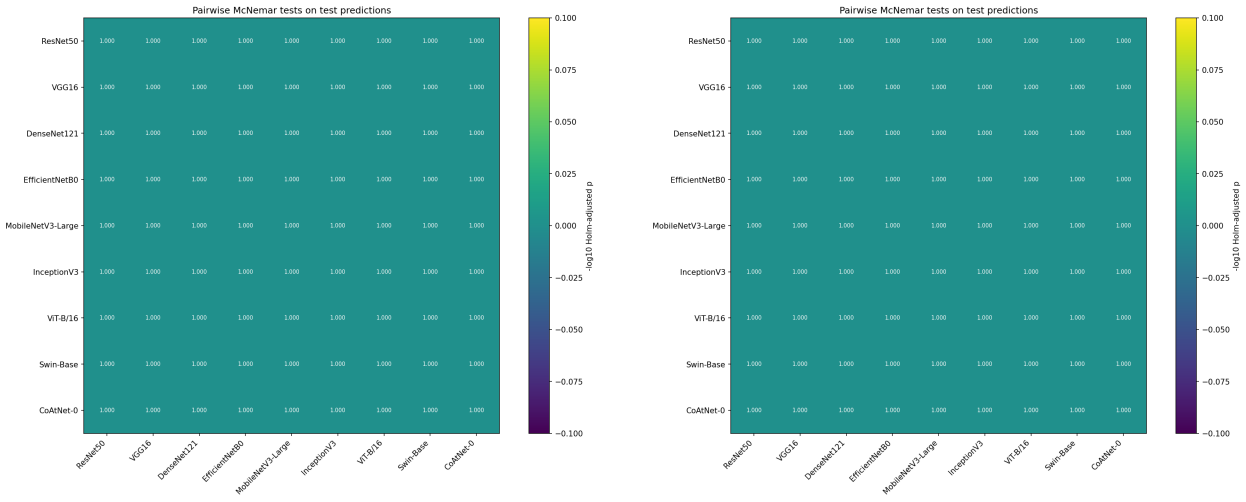
(a) Binary detection, tumour class.



(b) Subtype classification, one-versus-rest.

Figure S10: Receiver operating characteristic and precision–recall curves of every fold ensemble on the test set.

The Holm-corrected (Holm, 1979) McNemar (McNemar, 1947) p-values are shown in Figure S11, and test accuracy by distance to the development pool in Table S7 and Figure S12.

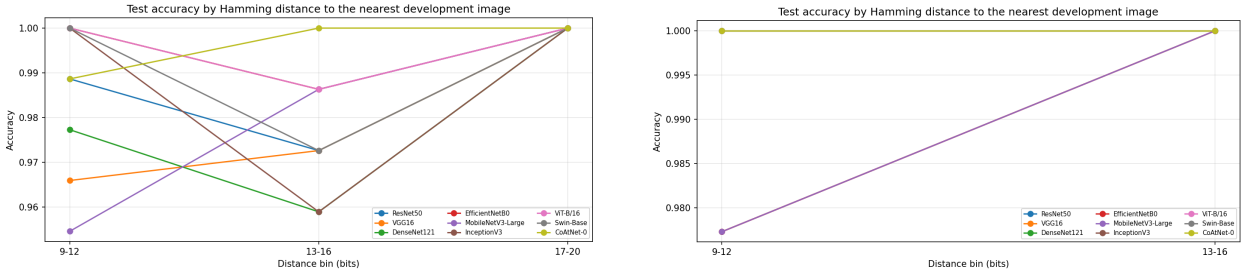


(a) Binary detection. (b) Subtype classification.

Figure S11: Holm-corrected exact McNemar p-values between the fold ensembles. Every adjusted p-value is 1.0 in both tasks.

Table S7
Fold-ensemble test accuracy by the Hamming distance from each test image to its nearest development image. The subtype classification test set has no image at distance 17 to 20.

	Binary detection			Subtype classification	
Model	9–12	13–16	17–20	9–12	13–16
CoAtNet-0	0.9886	1.0000	1.0000	1.0000	1.0000
DenseNet121	0.9773	0.9589	1.0000	1.0000	1.0000
EfficientNetB0	1.0000	0.9863	1.0000	0.9773	1.0000
InceptionV3	1.0000	0.9589	1.0000	1.0000	1.0000
MobileNetV3-Large	0.9545	0.9863	1.0000	0.9773	1.0000
ResNet50	0.9886	0.9726	1.0000	1.0000	1.0000
Swin-Base	1.0000	0.9726	1.0000	1.0000	1.0000
VGG16	0.9659	0.9726	1.0000	0.9773	1.0000
ViT-B/16	1.0000	0.9863	1.0000	1.0000	1.0000



(a) Binary detection. (b) Subtype classification.

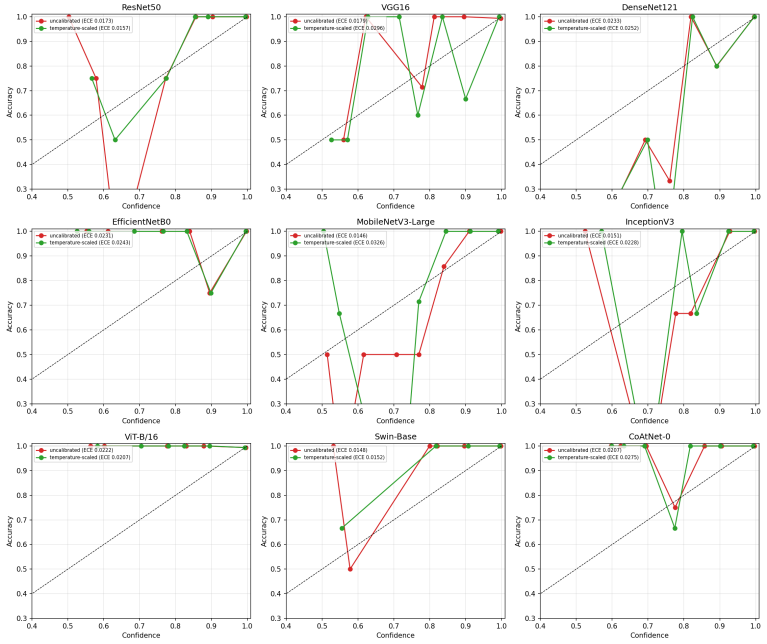
Figure S12: Test accuracy by the Hamming distance from each test image to its nearest development image.

At the mean validation-tuned Youden thresholds (Youden, 1950), which range from 0.43 to 0.84, binary detection accuracy is between 96.49% and 98.83%, and for EfficientNetB0 sensitivity falls from 100% to 96.26% while specificity

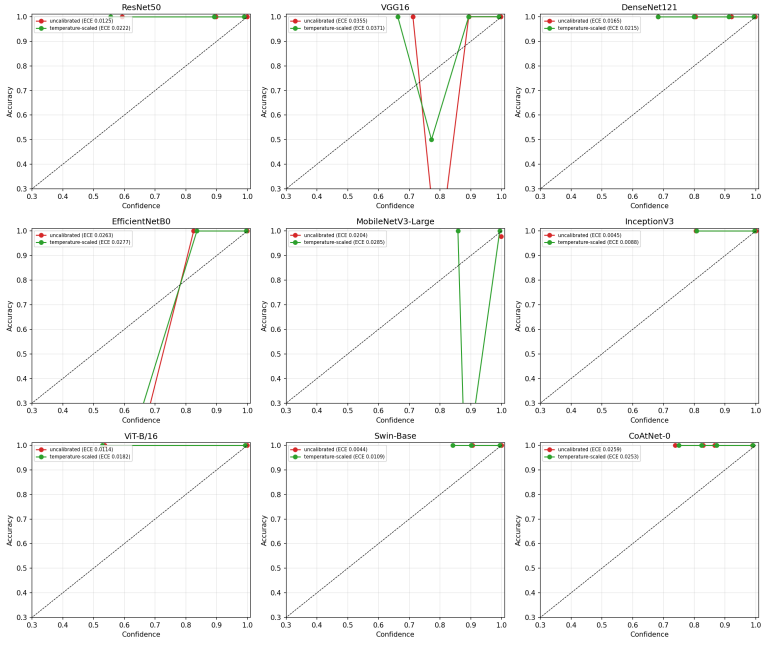
stays at 98.44%. The tuned thresholds raise accuracy only for DenseNet121 (97.66% against 97.08%), leave it unchanged for InceptionV3 and VGG16, and lower it for the other six models.

S6. Calibration and Efficiency

Reliability diagrams, which plot accuracy against confidence in each confidence bin (Guo et al., 2017), are shown in Figure S13.



(a) Binary detection.



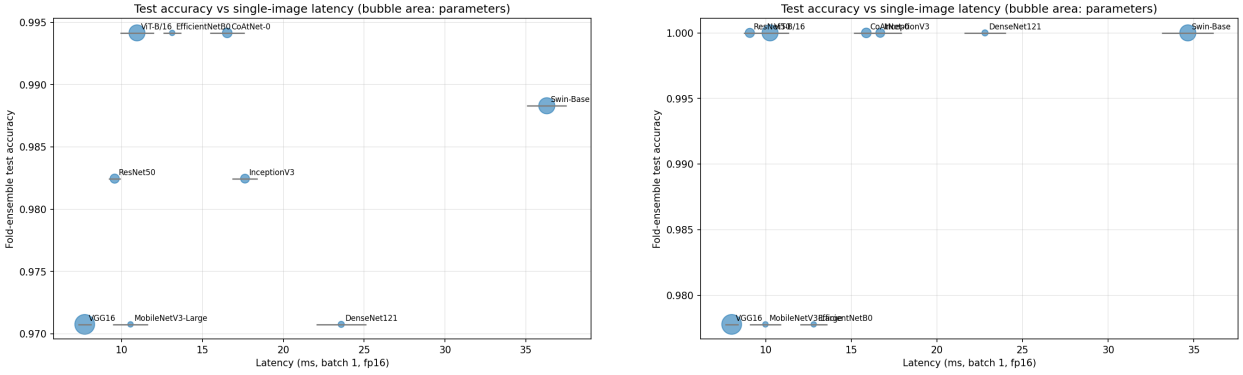
(b) Subtype classification.

Figure S13: Reliability diagrams of every fold ensemble on the test set, uncalibrated and temperature-scaled, with the expected calibration error of each.

Table S8 summarises the cost of the nine backbones, including GFLOPs and training time, and Figure S14 plots accuracy against latency. Peak training memory ranges from 2.1 GB for MobileNetV3-Large to 11.4 GB for CoAtNet-0.

Table S8
Computational cost of the nine backbones. Parameters and GFLOPs are measured on the binary detection models at the native input (299×299 for InceptionV3, which needs 5.67 GFLOPs at 224×224). Latency is the mean of single-image passes and throughput the mean at batch size 32, both on one Tesla T4 under half-precision autocasting and pooled over three repeats. Training time is minutes per run, averaged over folds. Rows are ordered by parameter count.

			Latency (ms)		Throughput (img/s)		Training (min/run)	
Model	Params (M)	GFLOPs	Bin.	Sub.	Bin.	Sub.	Bin.	Sub.
EfficientNetB0	4.01	0.77	13.1	12.8	1,393	1,266	2.0	5.1
MobileNetV3-Large	4.20	0.43	10.6	10.0	2,301	2,183	1.5	3.5
DenseNet121	6.95	5.67	23.6	22.8	657	608	1.5	6.9
InceptionV3	21.79	11.42	17.6	16.7	512	438	1.7	8.0
ResNet50	23.51	8.17	9.6	9.1	777	717	1.2	5.3
CoAtNet-0	26.67	8.42	16.5	15.9	513	448	2.3	9.5
ViT-B/16	85.80	22.54	11.0	10.3	272	241	2.9	15.0
Swin-Base	86.74	30.86	36.3	34.6	198	173	4.3	21.4
VGG16	134.26	30.93	7.7	8.0	386	332	2.2	9.5



(a) Binary detection.

(b) Subtype classification.

Figure S14: Fold-ensemble test accuracy against single-image latency for every model. Bin. and Sub. in Table S8 denote binary detection and subtype classification.

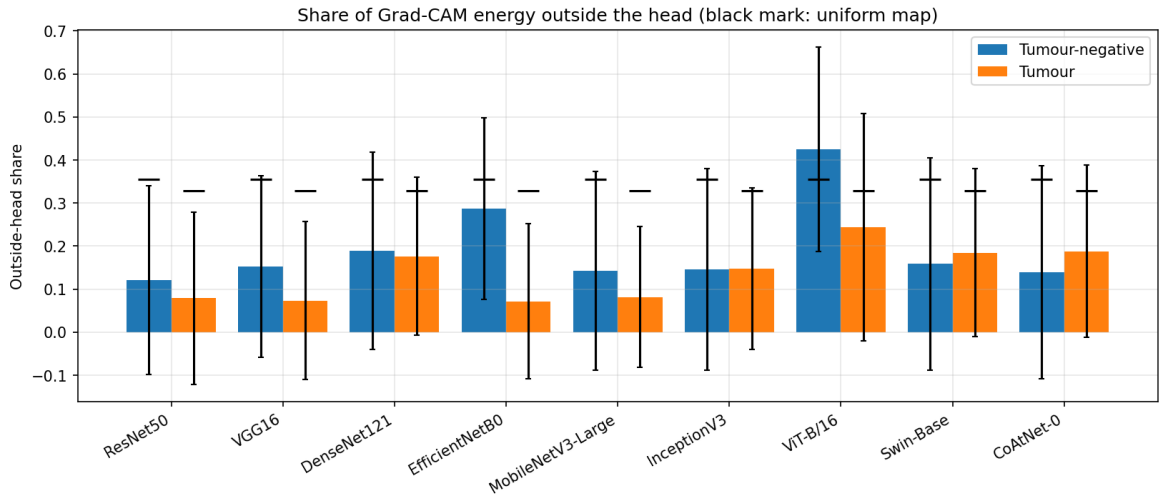
S7. Interpretability and Shortcut Details

The per-class shares of Grad-CAM (Selvaraju et al., 2017) energy and of occlusion sensitivity (Zeiler and Fergus, 2014) outside the head are listed in Table S9.

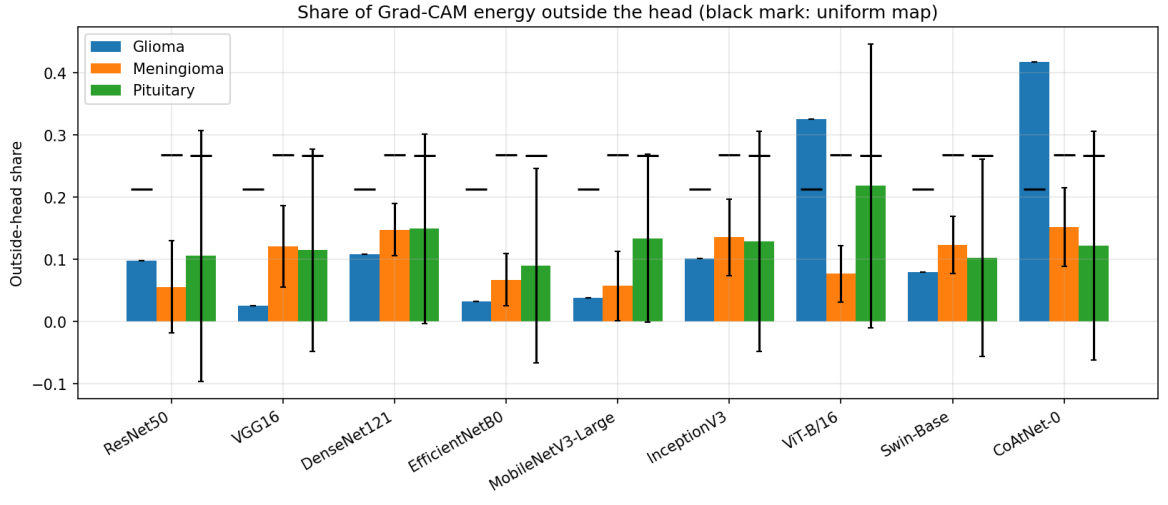
Table S9
Share of Grad-CAM energy outside the head mask, mean ± standard deviation over test images, and of occlusion sensitivity over 16 binary detection and 17 subtype classification test images. The subtype classification test set has one glioma image, so that column has no standard deviation.

	Binary detection			Subtype classification			
Model	Tumour-negative	Tumour	Occlusion	Glioma	Meningioma	Pituitary	Occlusion
ResNet50	0.122 ± 0.219	0.079 ± 0.200	0.116	0.098	0.056 ± 0.074	0.106 ± 0.202	0.079
VGG16	0.153 ± 0.212	0.074 ± 0.184	0.007	0.026	0.121 ± 0.066	0.115 ± 0.163	0.079
DenseNet121	0.190 ± 0.229	0.176 ± 0.183	0.135	0.108	0.148 ± 0.042	0.150 ± 0.153	0.110
EfficientNetB0	0.287 ± 0.211	0.072 ± 0.180	0.194	0.033	0.067 ± 0.042	0.091 ± 0.156	0.066
MobileNetV3-Large	0.142 ± 0.231	0.082 ± 0.163	0.111	0.038	0.058 ± 0.056	0.134 ± 0.136	0.071
InceptionV3	0.146 ± 0.234	0.147 ± 0.187	0.167	0.102	0.136 ± 0.062	0.129 ± 0.178	0.099
ViT-B/16 (Grad-CAM)	0.426 ± 0.238	0.244 ± 0.264	0.155	0.326	0.077 ± 0.045	0.218 ± 0.229	0.107
ViT-B/16 (rollout)	0.421 ± 0.179	0.285 ± 0.163		0.262	0.350 ± 0.047	0.358 ± 0.121	
Swin-Base	0.159 ± 0.246	0.185 ± 0.195	0.182	0.079	0.123 ± 0.046	0.103 ± 0.158	0.122
CoAtNet-0	0.139 ± 0.248	0.188 ± 0.200	0.089	0.417	0.152 ± 0.064	0.122 ± 0.184	0.124
Uniform map	0.356	0.328		0.213	0.268	0.267	

Figure S15 plots the same Grad-CAM shares per model and class against the share of a uniform map.



(a) Binary detection.



(b) Subtype classification.

Figure S15: Share of Grad-CAM energy outside the head per model and class; the black mark is the share of a uniform map.

The binary detection occlusion maps of every model are shown in Figure S16.

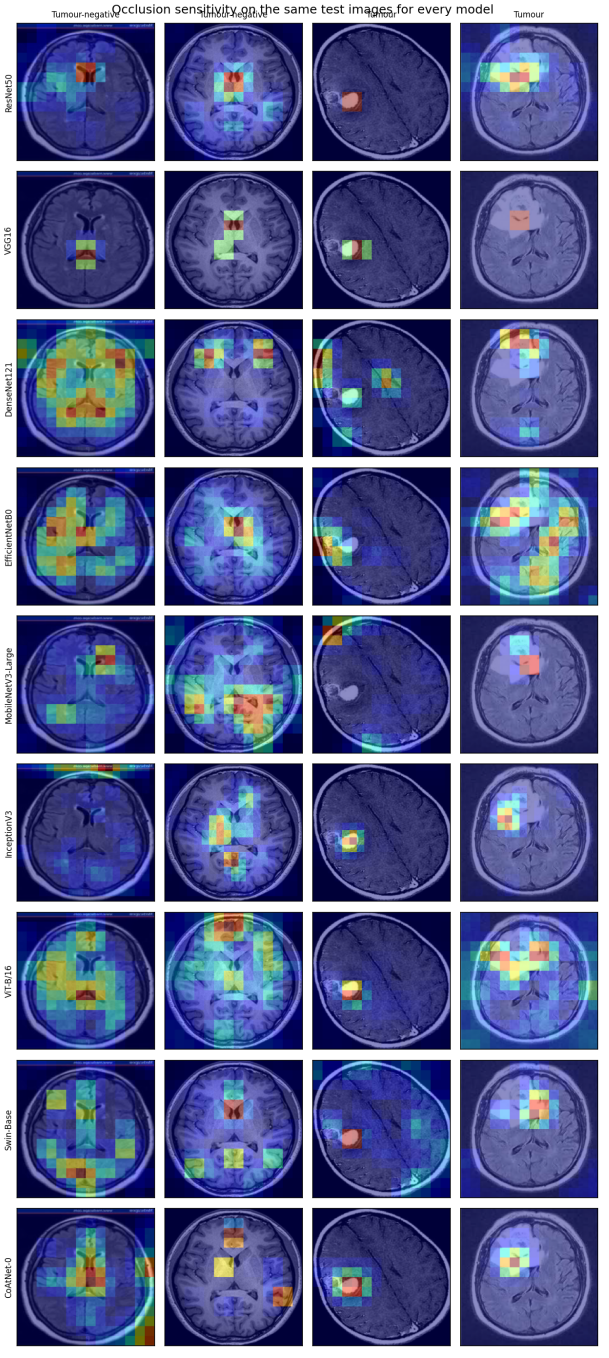


Figure S16: Binary detection: occlusion maps of every model on the same stratified test images.

The subtype classification occlusion maps of every model are shown in Figure S17.

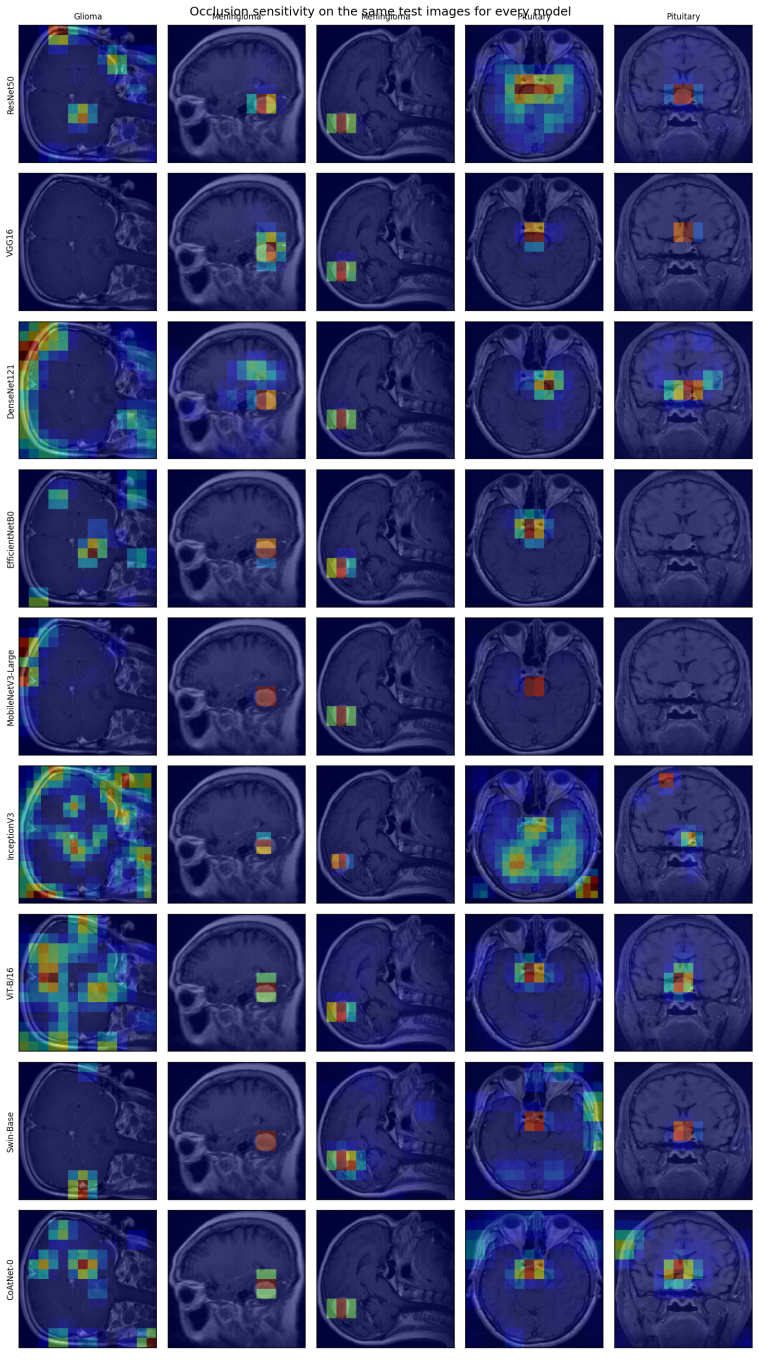
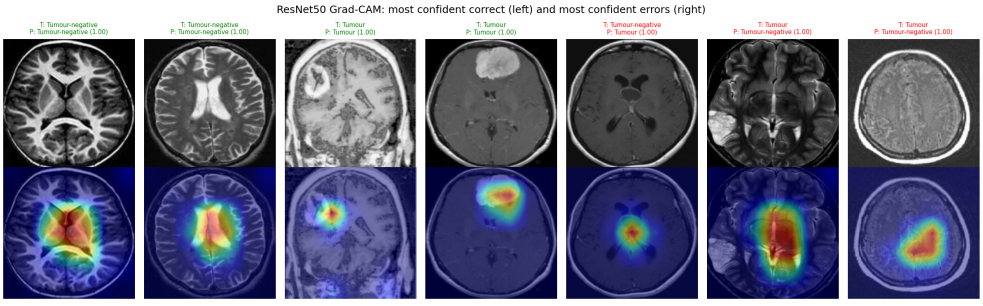
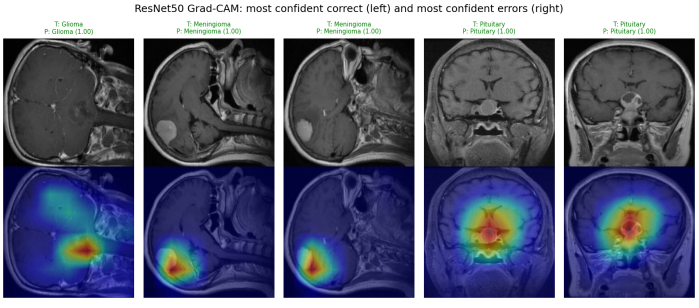


Figure S17: Subtype classification: occlusion maps of every model on the same stratified test images.

The Grad-CAM maps of the eight backbones not shown in the paper follow in the layout of Figure effig:gradcam of the paper: ResNet50 and VGG16 in Figures S18 and S19.

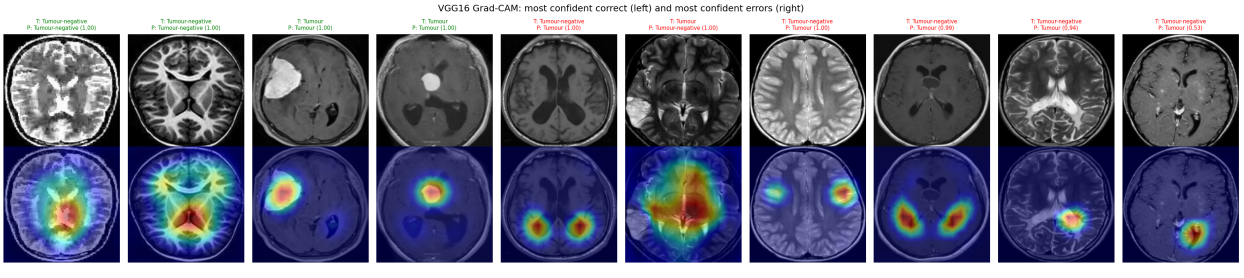


(a) Binary detection.

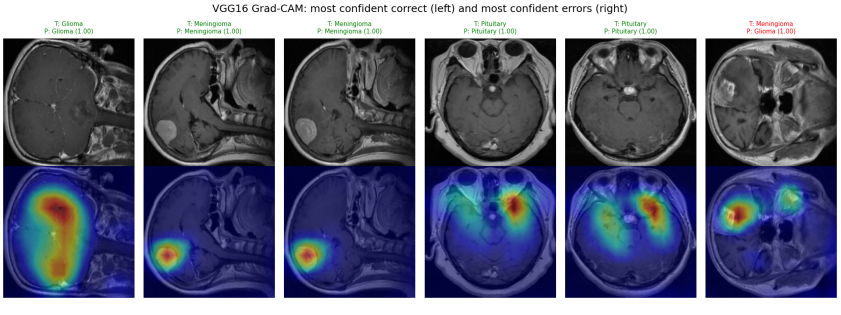


(b) Subtype classification.

Figure S18: Grad-CAM maps of ResNet50 on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.



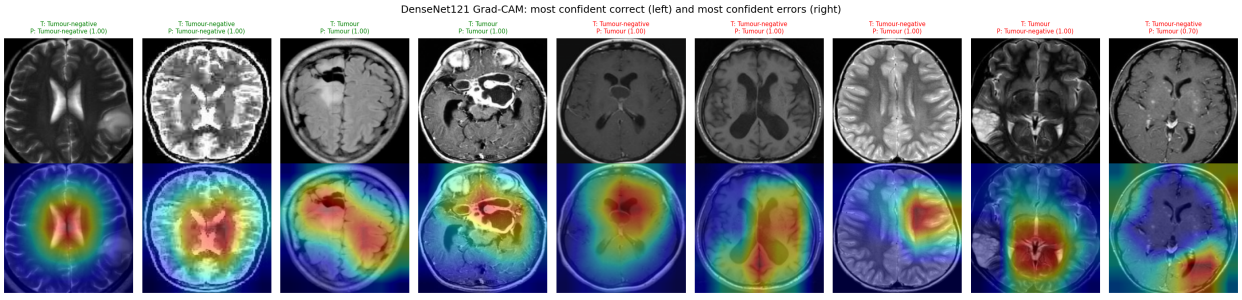
(a) Binary detection.



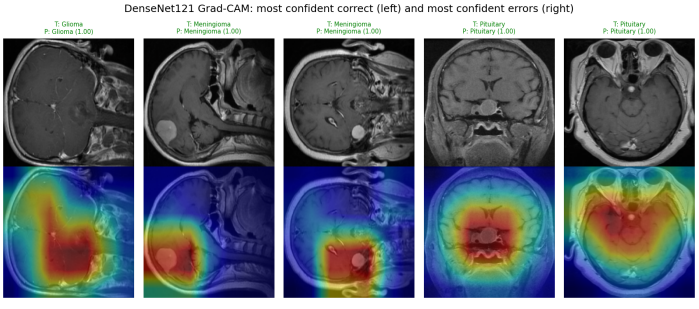
(b) Subtype classification.

Figure S19: Grad-CAM maps of VGG16 on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.

The maps of DenseNet121 and MobileNetV3-Large are shown in Figures S20 and S21.

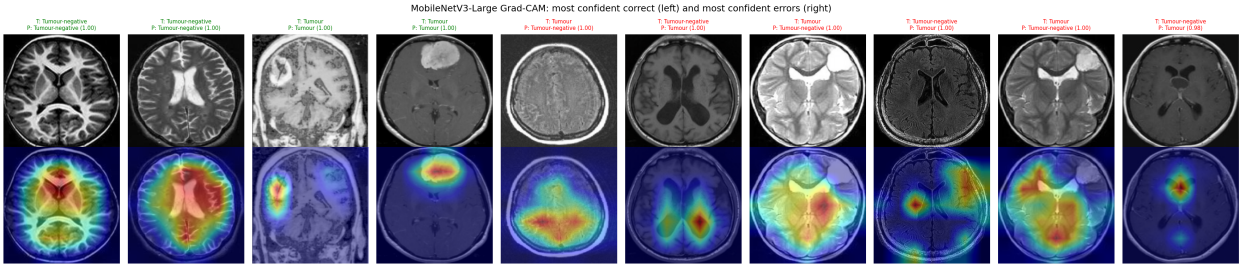


(a) Binary detection.

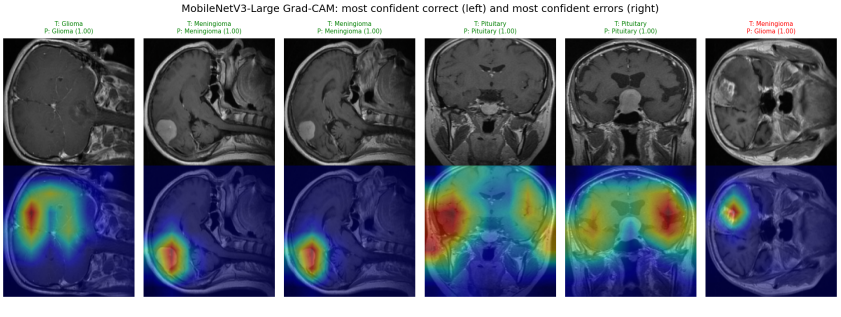


(b) Subtype classification.

Figure S20: Grad-CAM maps of DenseNet121 on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.



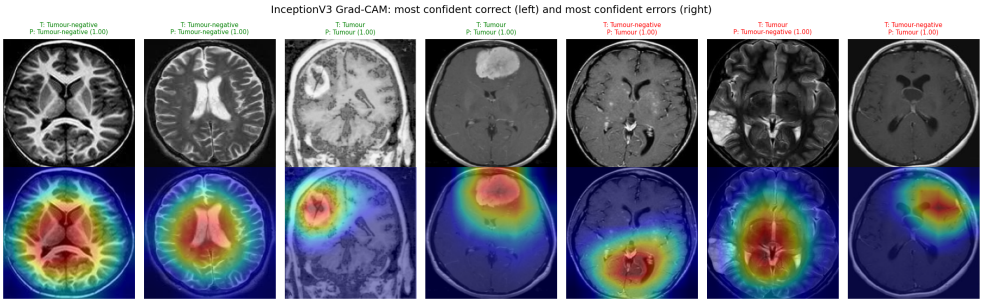
(a) Binary detection.



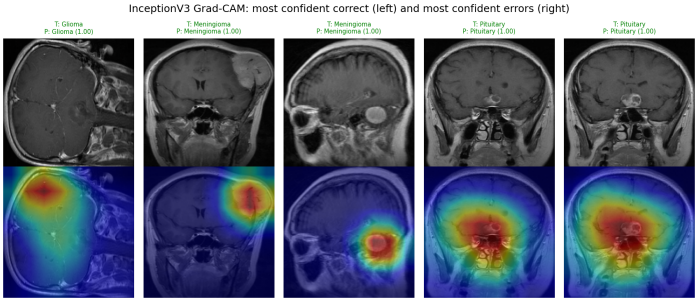
(b) Subtype classification.

Figure S21: Grad-CAM maps of MobileNetV3-Large on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.

The maps of InceptionV3 and ViT-B/16 are shown in Figures S22 and S23.

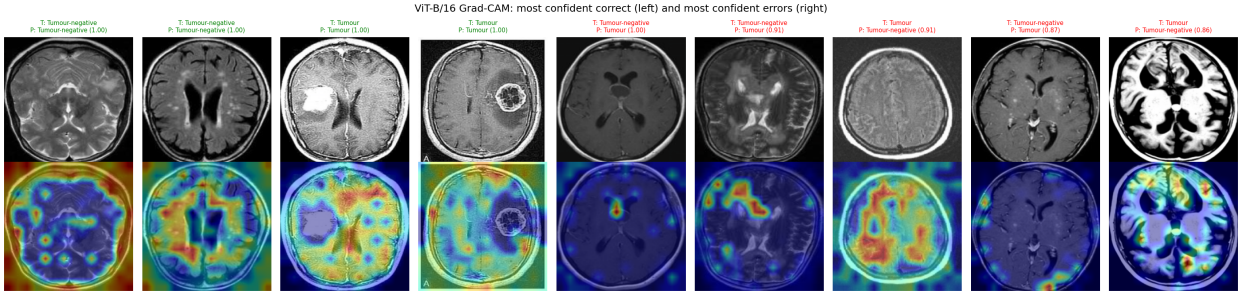


(a) Binary detection.

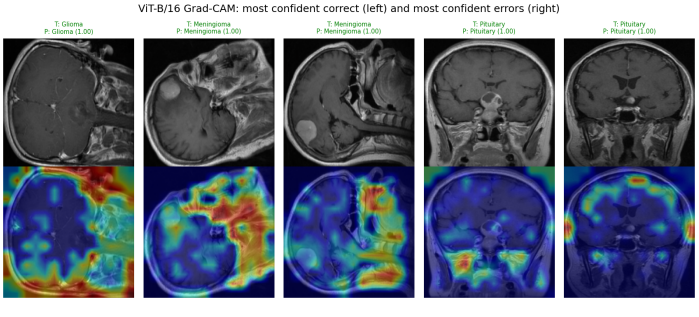


(b) Subtype classification.

Figure S22: Grad-CAM maps of InceptionV3 on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.



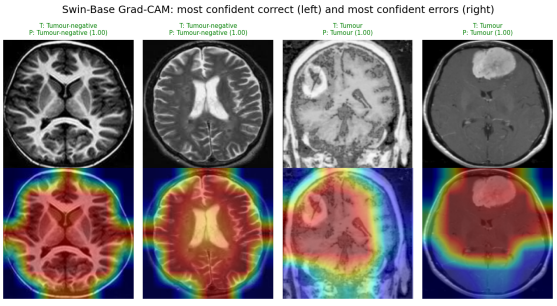
(a) Binary detection.



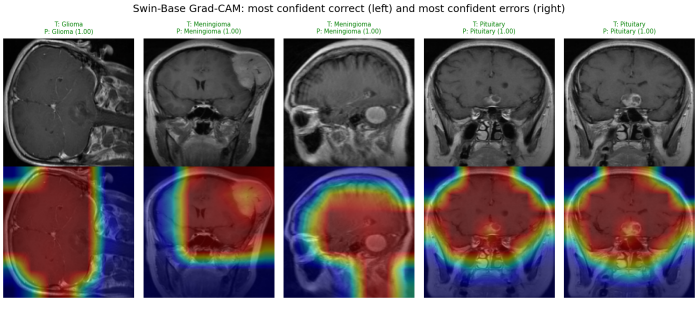
(b) Subtype classification.

Figure S23: Grad-CAM maps of ViT-B/16 on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.

The maps of Swin-Base and CoAtNet-0 are shown in Figures S24 and S25.

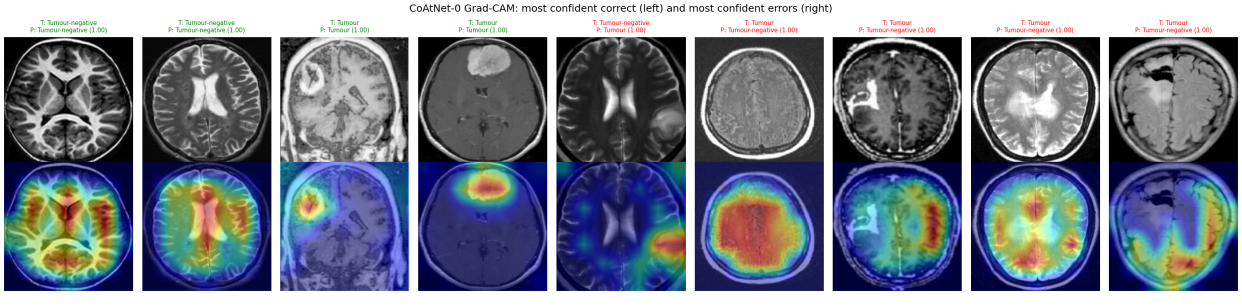


(a) Binary detection.

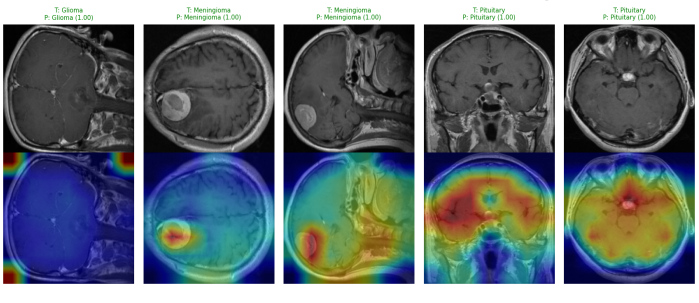


(b) Subtype classification.

Figure S24: Grad-CAM maps of Swin-Base on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.



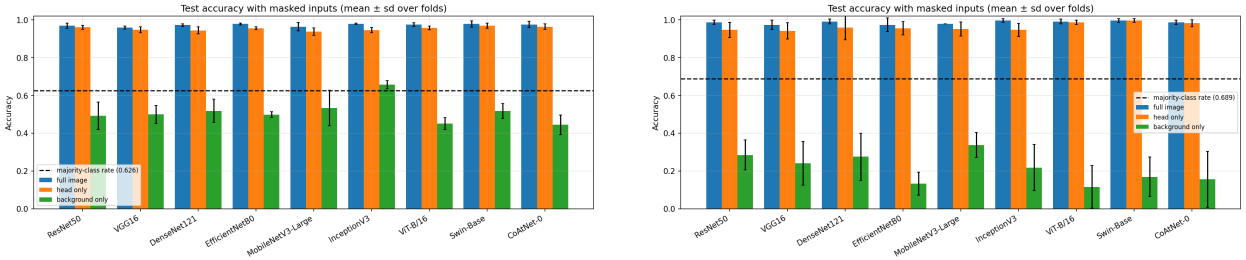
(a) Binary detection.



(b) Subtype classification.

Figure S25: Grad-CAM maps of CoAtNet-0 on its most confident correct and most confident wrong test predictions, from the fold with the lowest validation loss.

The masking test results are shown in Figure S26.



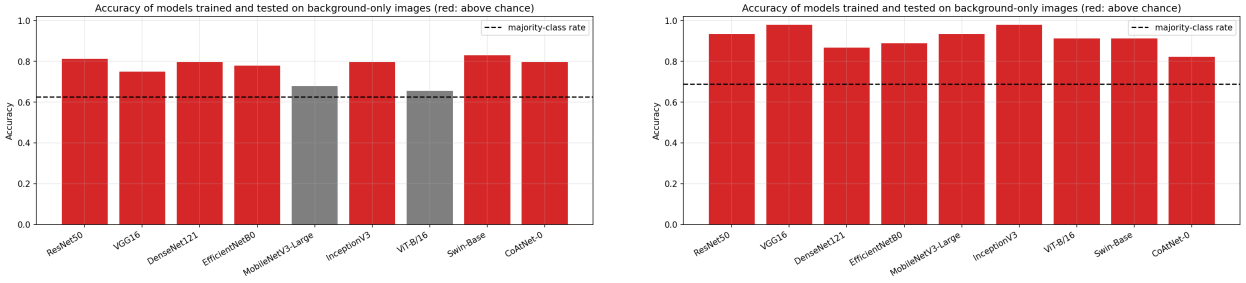
(a) Binary detection. (b) Subtype classification.

Figure S26: Masking test (inference only): test accuracy of each fold's checkpoint on full images, head-only images (background set to zero) and background-only images (head set to zero), mean ± standard deviation over folds.

The shortcut test (Geirhos et al., 2020) results are listed in Table S10 and plotted in Figure S27. Background-only AUCs reach 0.889 in binary detection and lie between 0.972 and 1.000 in subtype classification.

Table S10
Shortcut test: models trained (fold 0) and tested on background-only images. The p-value is a one-sided exact binomial test against the majority-class rate (0.6257 binary detection, 0.6889 subtype classification), Holm-corrected within each task.

	Binary detection			Subtype classification		
Model	Accuracy	AUC	Holm p	Accuracy	AUC	Holm p
ResNet50	0.8129	0.8657	$< 10^{-4}$	0.9333	0.9968	0.0006
VGG16	0.7485	0.8359	0.0013	0.9778	1.0000	$< 10^{-4}$
DenseNet121	0.7953	0.8578	$< 10^{-4}$	0.8667	0.9904	0.0104
EfficientNetB0	0.7778	0.8584	0.0001	0.8889	1.0000	0.0048
MobileNetV3-Large	0.6784	0.7821	0.1772	0.9333	1.0000	0.0006
InceptionV3	0.7953	0.8606	$< 10^{-4}$	0.9778	0.9984	$< 10^{-4}$
ViT-B/16	0.6550	0.7411	0.2396	0.9111	0.9717	0.0020
Swin-Base	0.8304	0.8893	$< 10^{-4}$	0.9111	0.9976	0.0020
CoAtNet-0	0.7953	0.8756	$< 10^{-4}$	0.8222	0.9752	0.0337



(a) Binary detection. (b) Subtype classification.

Figure S27: Shortcut test: accuracy on background-only test images of models trained on background-only images (fold 0); red bars are significantly above the majority-class rate after Holm correction.

References

Bhuvaji, S., Kadam, A., Bhumkar, P., Dedge, S., Kanchan, S., 2020. Brain tumor classification (MRI). Kaggle dataset. doi:10.34740/KAGGLE/DSV/1183165.

Chakrabarty, N., 2019. Brain MRI images for brain tumor detection. Kaggle dataset, version 1. URL: https://www.kaggle.com/datasets/navoneel/brain-mri-images-for-brain-tumor-detection.

Cheng, J., 2017. Brain tumor dataset. figshare, version 5. doi:10.6084/m9.figshare.1512427.v5.

Cheng, J., Huang, W., Cao, S., Yang, R., Yang, W., Yun, Z., Wang, Z., Feng, Q., 2015. Enhanced performance of brain tumor classification via tumor region augmentation and partition. PLoS ONE 10, e0140381. doi:10.1371/journal.pone.0140381.

Deng, J., Dong, W., Socher, R., Li, L.J., Li, K., Fei-Fei, L., 2009. ImageNet: A large-scale hierarchical image database, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 248–255. doi:10.1109/CVPR.2009.5206848.

Efron, B., 1979. Bootstrap methods: Another look at the jackknife. The Annals of Statistics 7, 1–26. doi:10.1214/aos/1176344552.

Fawcett, T., 2006. An introduction to ROC analysis. Pattern Recognition Letters 27, 861–874. doi:10.1016/j.patrec.2005.10.010.

Geirhos, R., Jacobsen, J.H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., Wichmann, F.A., 2020. Shortcut learning in deep neural networks. Nature Machine Intelligence 2, 665–673. doi:10.1038/s42256-020-00257-z.

Gneiting, T., Raftery, A.E., 2007. Strictly proper scoring rules, prediction, and estimation. Journal of the American Statistical Association 102, 359–378. doi:10.1198/016214506000001437.

Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q., 2017. On calibration of modern neural networks, in: Proceedings of the 34th International Conference on Machine Learning (ICML), pp. 1321–1330. URL: https://proceedings.mlr.press/v70/guo17a.html.

Hamada, A., 2025. Br35H :: Brain tumor detection 2020. IEEE DataPort; also available at https://www.kaggle.com/datasets/ahmedhamada0/brain-tumor-detection. doi:10.21227/tbkk-q937.

Holm, S., 1979. A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics 6, 65–70. URL: https://www.jstor.org/stable/4615733.

Loshchilov, I., Hutter, F., 2017. SGDR: Stochastic gradient descent with warm restarts. International Conference on Learning Representations (ICLR). URL: https://mlanthology.org/iclr/2017/loshchilov2017iclr-sgdr/.

Loshchilov, I., Hutter, F., 2019. Decoupled weight decay regularization. International Conference on Learning Representations (ICLR). URL: https://mlanthology.org/iclr/2019/loshchilov2019iclr-decoupled/.

McNemar, Q., 1947. Note on the sampling error of the difference between correlated proportions or percentages. Psychometrika 12, 153–157. doi:10.1007/BF02295996.

Micikevicius, P., Narang, S., Alben, J., Diamos, G., Elsen, E., Garcia, D., Ginsburg, B., Houston, M., Kuchaiev, O., Venkatesh, G., Wu, H., 2018. Mixed precision training. International Conference on Learning Representations (ICLR). URL: https://mlanthology.org/iclr/2018/micikevicius2018iclr-mixed/.

Pascanu, R., Mikolov, T., Bengio, Y., 2013. On the difficulty of training recurrent neural networks, in: Proceedings of the 30th International Conference on Machine Learning (ICML), pp. 1310–1318. URL: https://proceedings.mlr.press/v28/pascanu13.html.

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Kopf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., Bai, J., Chintala, S., 2019. PyTorch: An imperative style, high-performance deep learning library. Advances in Neural Information Processing Systems (NeurIPS) 32. URL: https://proceedings.neurips.cc/paper/2019/hash/bdbca288fee7f92f2bfa9f7012727740-Abstract.html.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, É., 2011. Scikit-learn: Machine learning in Python. Journal of Machine Learning Research 12, 2825–2830. URL: https://jmlr.org/papers/v12/pedregosa11a.html.

Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D., 2017. Grad-CAM: Visual explanations from deep networks via gradient-based localization, in: Proceedings of the IEEE International Conference on Computer Vision (ICCV), pp. 618–626. doi:10.1109/ICCV.2017.74.

Stone, M., 1974. Cross-validatory choice and assessment of statistical predictions. Journal of the Royal Statistical Society: Series B (Methodological) 36, 111–133. doi:10.1111/j.2517-6161.1974.tb00994.x.

Youden, W.J., 1950. Index for rating diagnostic tests. Cancer 3, 32–35. doi:10.1002/1097-0142(1950)3:1<32::AID-CNCR2820030106>3.0.CO;2-3.

Zeiler, M.D., Fergus, R., 2014. Visualizing and understanding convolutional networks, in: Computer Vision – ECCV 2014, pp. 818–833. doi:10.1007/978-3-319-10590-1_53.

Zitová, B., Flusser, J., 2003. Image registration methods: A survey. Image and Vision Computing 21, 977–1000. doi:10.1016/S0262-8856(03)00137-9.